case report

DIABETES MELLITUS AND INSULIN IN AN ASPIRIN SENSITIVE ASTHMATIC

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BECAUSE DIABETES mellitus and asthma seldom occur together in an individual[1,2] and because aggravation of asthma by insulin has not previously been described, the following case is deemed worthy of reporting. The nonatopic mechanism of this aggravation of asthma by insulin is explored. This patient's reaction stimulated an examination of the effect of insulin upon the cyclic adenosine monophosphate (CAMP) and cyclic guanosine monophosphate (GMP) level and mediator release in mast cells which has not previously been investigated.

The common occurrence of yellow dye #5, tartrazine, in many common antiasthmatic preparations is noted.[3] Tartrazine, like aspirin, commonly produces severe asthma in aspirin sensitive asthmatics.[4,5,6]

Case Report

R.L., a 45-year-old white male, was first seen on March 26, 1975, because of severe asthma of six years duration.

Past History

Except for surgery to correct a broken nose in his late teens and a tonsillectomy at age 21 for recurrent tonsillitis, this patient had good health in his first two and one-half decades. At age 25 severe diabetes mellitus developed which required dietary restrictions and large doses of insulin up to 80 or more units per day. A herniorrhaphy was performed in 1946. In September of 1969 a bilateral nasal polypectomy was done. A vasectomy was performed in 1970.

Family History

The patient's father suffered from severe asthma, type undetermined, from age 18 to 38. His asthma spontaneously remitted at age 38, never to return. A maternal uncle developed diabetes four years before the date of this report. The family history is otherwise negative for communicable, infectious, malignant, or hereditary disease.

Laboratory Data

R.B.C. —5,000,000
Hemoglobin —14.4 gms.
Hematocrit —42.1%
Mean corpuscular volume —84%
Mean corpuscular hemoglobin —28.7%
White blood count —10,400
Polymorph nuclear leucocytes —54%
Lymphocytes —38%

Eosinophils —7%
Monocytes —1%
Urinalysis — Specific gravity — 1.020
Albumin —negative
Glucose —negative
Miscroscopic —negative
Reaction —acid
Calcium —9.8 mgm%
Repeated blood glucose — 97-320 mgm
Blood urea nitrogen —5.2 mgm%
Albumin —3.7 gms%
Total protein —5.9 gms%
Bilirubin —0.4 mgm%
Alkaline phosphatase — 384 units per cc
L.D.H. —2204 units per cc
S.G.O.T. —304 units per cc
Cholesterol —258 mgm%
Triglycerides —200 mgm%
Immunoglobulins: (by radial immuno-diffussion)
IgA —20 mgm%
IgG —435 mgm%
IgM —177 mgm%

Electrocardiogram: Normal with right axis deviation.

A chest X-ray was normal. X-ray of the cervical spine revealed early degenerative changes in the fifth and sixth cervical vertebrae. Intradermal skin testing with 1-1000 V-80 semilente insulin, ultralente and regular insulin were negative. After skin tests to insulin, Bruce Petersen, Ph.D., of the Eli Lilly Research Department, performed the following antibody studies on the patient's serum:

IgG and anti-insulin antibody titers were determined using a procedure similar to that described by Wright et al.[7] Briefly, I^{125} pork and I^{125} Insulin is incubated with several dilutions of the serum sample for 30 minutes. Cellulose powder is then added and after 30 minutes the mixture is centrifuged. The supernatant is then counted on a gamma counter and the result is compared with that found with normal (non-diabetic) serum and to high-titer guinea pig anti-serum. The results are expressed in units. Normal ranges are 0-15 units.

IgE anti-insulin antibodies were detected using the radio-immunosorbent technique of Wide et al.[8] IgE antibodies absorbed to insulin-coated sepharose 4B particles (pharmacia chem.) are detected using I^{125} labeled anti-human IgE antisera. The test is positive if the uptake of I^{125} counts is two to five times that of the control.

Results:	Units	
Anti-Insulin	Beef	Pork
IgG	8.5	5.8
	Cts/min.	
IgE	Beef	Pork
Control	3062	2478
R.L.	4265	6289
Ratio	2	2,54
R.L./C	—slightly above normal no RmAC (0-2)	

Summary

The assays done in our laboratories did not detect the presence of IgG anti-insulin antibodies in the serum of R.L. A small amount of IgE anti-pork, but not anti-beef-insulin antibodies, were detected in the patient's serum. It is not unreasonable to expect low levels of IgE anti-insulin antibodies to be present in the serum of an allergy-prone person who has received insulin over a period of years. The presence on the patient's serum of IgE anti-insulin antibodies does not provide evidence that there is a relationship between his asthmatic condition and insulin reaginic antibodies.

Present Illness

His present illness started in October, 1969, one month after his nasal polypectomy. Asthma started at that time and has persisted perennially every day and night to the present. In 1971 aspirin for a headache was followed in 25 minutes by severe asthma. Again in 1973 aspirin given for a headache was followed by severe asthma. He has been taking cortisone continuously since 1971 in an effort to control his severe asthma. This has aggravated his diabetes and necessitated larger doses of insulin.

He was twice before evaluated on an allergic basis. After his first evaluation in 1969 he was told that he was allergic to bacteria. He received injections of bacterial vaccine for two years without aid. After his second allergic evaluation in 1971 he was treated with injections of dust and molds for two years, again without relief. He was bronchoscoped in 1970 and again in 1971. An autogenous vaccine grown from the bronchoscopic aspirations was given for two years without any change in the severity of his asthma. Bronchoscopic washings were negative for malignant cells.

Psychiatric evaluation was requested in early 1974 to determine whether his severe asthma might not be on an emotional basis. His psychiatrist stated: "I am firmly convinced that emotional conflict is not the cause of his asthma." He urged his physicians to continue to treat him on the basis of organic, not emotional, disease. He was receiving the following medications:

1. Marax tablets — 4 daily
2. Atromid-S — 4 daily for hyperlipemia
3. Erythromycin Estolate (Lilly's Ilosone) — mgm 250 — 4 daily
4. Digitoxin — mgm 0.1 daily
5. Enteric tablets potassium iodide, gr.V (Lilly's Enseals) — 3 daily
6. Micebrin T. (Lilly) — 1 daily
7. Vitamin C — 1000 mgm daily
8. Paramethasone Acetate (Haldrone, Eli Lilly) — mgm 6.00 every other day
9. Semilente Insulin, 18 units mixed with Ultralente Insulin — 24 units every a.m.
10. Regular insulin — units 40 at 5:00 p.m. every evening.

He had also taken sodium cromolyn in the past without relief. In addition he was using a Bennett machine with Bronkosol every night and usually two to four times per night to relieve his severe paroxysms. His breathing was so labored and noisy that he slept in a bedroom of his own so as not to disturb other members of the household. He also complained of the voluminous amount of mucus his lungs produced each day. (It is to be noted that when the Bennett machine and Bronkosol were discontinued this voluminous production of mucus ceased.) During the past six years following insulin shock reactions his asthma abated for four to six hours.

Physical Examination

Physical examination revealed large bilateral nasal polyps and severe asthma. His blood pressure was 180/100. Intradermal skin tests to common household allergens, pollens and molds were entirely negative. Scratch and intradermal skin tests to 1-1000 dilution of Semilente, Ultralente and regular insulin were also negative. His skin did give a good four-plus response to testing with a 1-10,000 solution of histamine.

Checking with the manufacturers, it was found that Haldrone, Ilosone Capsules, and Enseals of Potassium Iodide all contained tartrazine. Atromid-S, according to correspondence from the manufacturers, sometimes was prepared with tartrazine, which aggravates the aspirin sensitive asthmatic. He was started on the following regimen:

1. He was asked to discontinue the Bennett machine and Bronkosol inhalations.

2. Marax was continued four times daily.

3. Prednisone, mgm 15.00 every other day, was substituted for Haldrone.

4. Saturated solution of potassium iodide, 10 drops four times daily, was substituted for the Enseals of Potassium Iodide.

5. Susphrine (1-200) 0.2 cc was to be administered every six hours as needed for relief.

He continued to have severe asthma. His diabetes was worse on the day he took Prednisone. His prednisone was gradually increased to mgm 40 every a.m. within one month. Although improved, he continued to awaken each night with severe asthma and required Susphrine once or twice every night. In spite of this he stated he was markedly improved and no longer produced voluminous quantities of mucus as he had done while on the Bennett machine and Bronkosol inhalations.

Much thought was given as to why this aspirin sensitive asthmatic with nasal polyps should be so much more difficult to manage even after aggravating factors such as the Bennett machine and Bronkosol and drugs containing tartrazine had been eliminated. It became obvious that somehow his diabetes might be complicating the picture. Epinephrine not only raises blood sugar levels but also reduces bronchial edema and bronchospasm. Could it be that insulin is an antagonist since it reduces the blood sugar and perhaps increases bronchial edema and bronchospasm as well? His evening 40 units of regular insulin were discontinued. Acetohexamide (Eli Lilly's Dymelor), an oral antidiabetic agent, was substituted for the insulin; 1500 mgm were given every evening. Although his diabetes is not as well controlled, he no longer awakens with asthma. He has discontinued the use of Susphrine. Prednisone has been reduced to mgm 2.5 twice daily. Physical examination now reveals normal, quiet breathing and no asthma. His blood pressure was 122/80. He now plays golf and tennis regularly, which he had not been able to do since the onset of his asthma. His breathing is again quiet and peaceful and his wife has been able to move back into the bedroom with him.

Discussion

The infrequent occurrence of asthma among diabetics and of diabetes among asthmatics has long been known to allergists and those who care for diabetics. Swern[1] observed only six diabetics among 4,000 allergic patients. He also noted that when diabetes did develop in allergic patients, their skin tests became less intense and their asthma improved. Again, could a deficiency of insulin be the factor? Konig[2] in 1935 found three asthmatics among 1,240 diabetics. This infrequent concurrence of the two diseases becomes quite evident when one realizes that the incidence of diabetes has been estimated by Joslin[9] to be from 2.5-5% of the adult population. Konig in 1935 suggested an antagonism between epinephrine and insulin as an explanation of this phenomenon. Walker is quoted by Joslin[10] as having seen two diabetics among 3,000 asthmatics.

COINCIDENCE OF ASTHMA AND DIABETES.

1. Walker (1928), 2 diabetics among 3,000 asthmatics.
2. Swern (1932), 6 diabetics among 4,000 allergic patients.
3. Konig (1935), 3 asthmatics among 1,240 diabetics.
4. Joslin (1971), 2.5% - 5% of adult population have diabetes.

Wegierko[11] treated 40 asthmatics with insulin shock which temporarily relieved their asthma. It is to be noted that this occurred in our patient, who was relieved of his asthma for four to six hours after accidental insulin shock. This type of therapy has never come into general use, probably because of the violence of this type of therapy (which had to be followed by intravenous glucose) as compared to an injection of epinephrine.

Abrahamson[12] described 12 patients with what he termed hyperinsulinism in whom asthma developed when the blood sugar was very low (48-60 mgm per 100 cc). (See Table I) On a low carbohydrate, high fat diet and frequent feeding these patients improved. He stated that the high fat content of the diet suppressed the islet cells of the pancreas and reduced the release of insulin. This prevented the severe hypoglycemia which was concurrent in his patients with the development of attacks of asthma. He also stated that in asthmatics the occurrence of nocturnal asthma could be explained by this mechanism. Repeated glucose tolerance tests in four of these patients six months later were normal. (See Table II) This would certainly point toward their previous diagnosis of hyperinsulinism being secondary to an improper diet. Today we might explain this phenomenon on a less active beta adrenergic receptor system.

Abrahamson suggested a reciprocal relationship between asthma and diabetes, since asthmatic attacks tend to occur with hypoglycemia. He also noted that asthmatics are made worse by sodium chloride and diabetics improve. Asthmatics are worse with hypothyroidism and better with hyperthyroidism. The reverse is true in diabetes. (See Table III) All this seems to point to a reciprocal rela-

TABLE I. LOW BLOOD SUGAR AND ASTHMA.

	Patient	Fasting	Hours After Oral Glucose						
			1	2	3	4	5	6	7
Seasonal Hay fever and asthma	1	64*	132	97	82	70	56*		
	2	80	150	87	80	69	58*		
	3	110	144	118	96	84	73	66	55*
	4	64*	90	79	69	56*			
	5	118	152	120	102	83	72	64*	
	6	98	142	100	82	69	62	54*	
	7	98	138	106	78	61*			
	8	105	152	102	83	76	67	58	48*
	9	88	136	92	82	76	70	62*	
Perennial asthma	10	65	136	110	88	72	65	61*	
	11	125	144	109	86	76	58*		
	12	59*	110	94	83	71	65	57*	

*Patient developed asthma. Asthma relieved with milk.
From Abrahamson – Reference 11.

TABLE II. REPEATED GLUCOSE TOLERANCE ON FOUR PATIENTS AFTER HARRIS DIET.

Patient	Fasting	Hours After Oral Glucose					
		1	2	3	4	5	6
1	82	134	88	86	82	87	78
2	90	138	89	89	94	85	80
3	118	142	110	116	112	110	93
4	106	129	125	108	106	112	100

From Abrahamson – Reference 11.

tionship between asthma and diabetes as well as between epinephrine and insulin.

Long[13] noted that the ratio of insulin to cortisone output in guinea pigs profoundly influenced immunological and allergic reactions to infections. Fischel and Szentivanyi[14] demonstrated that additional insulin sensitized normal mice to the action of histamine and serotonin. Adamkiewicz et al[15,16] were able to produce anaphylactoid reactions to injections of dextran in animals given insulin who would not react in this manner without the addition of insulin. They were also able to demonstrate that in rats a single subcutaneous injection of 10 units of insulin allowed the development of anaphylactoid reactions to doses of dextran too small to shock the animals if given without the additional insulin. Sanyal[17] injected insulin intraperitoneally in animals who were simultaneously injected with egg white. The insulin increased the degree of anaphylactic shock but had no affect on antibody formation.

Insulin acts upon liver cells by decreasing cyclic adenosine monophosphate (CAMP).[18] The decreased CAMP level of the liver cells has an inhibitory affect and prevents the release of glucose from the cells into the general circulation. (See Figure 1) Epinephrine and glucagon produce glycogenolysis and gluconeogensis. This results in a release of glucose into the circulation from the liver cells. This if felt to be promulgated by an increased CAMP level within the liver cells. Again we see the antagonism between epinephrine and insulin.

Cerasi et al[19] added further impetus to the beta adrenergic receptor system and CAMP level of the cell as a factor in glucose control. Intravenous Propanolol, a beta-blocking agent, suppressed insulin release in 16 healthy subjects who were concurrently receiving infusions of glucose. They felt that glucose receptors are sensitive to changing cellular levels of CAMP. There is controversy on this concept, however. Robertson and Porte[20] stated that Propanolol abolished the response to

TABLE III. RECIPROCAL RELATIONSHIP

	Asthma	Diabetes
Blood sugar	Low	High
Sodium Chloride	Aggravates	Improves
Hypothyroidism	Aggravates	Improves
Hyperthyroidism	Improves	Aggravates
Epinephrine	Improves	Aggravates
Insulin	Aggravates	Improves

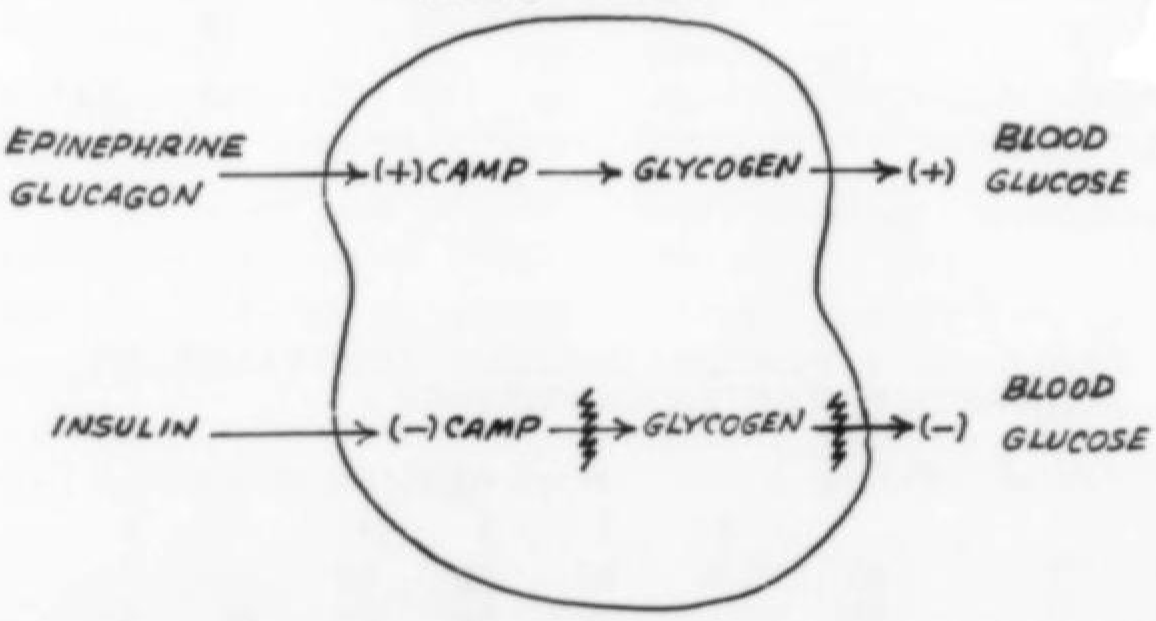

Figure 1. Epinephrine and/or Glucagon, acting via the adrenergic receptor, stimulate cyclic AMP inducing glycogenolysis and increased blood glucose. Insulin lowers cyclic AMP, thus blocking glycogenolysis and lowering blood glucose.

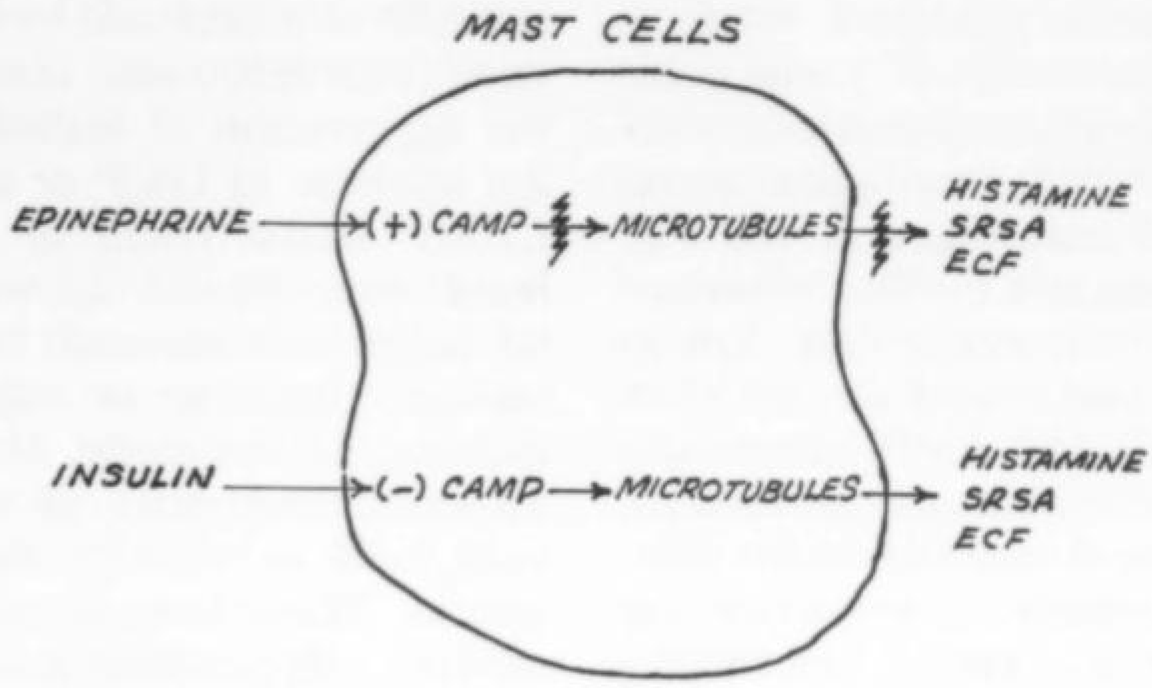

Figure 2. Epinephrine causes elevation of cyclic AMP, thus blocking microtubular release of chemical mediators. Insulin causes reduction in cyclic AMP, thus allowing microtubular release of chemical mediators.

isoproterenol but not to glucose. Hedstrand and Aberg[21] feel that the insulin response to glucose is independent of the beta adrenergic receptor system.

The beta-blockage theory of asthma as proposed by Szentivanyi[22] and the effect of the CAMP level in the mast cell has been brilliantly summarized by Austen and Lichtenstein.[23] An increase in the level of CAMP in the mast cells of the lung, as produced by epinephrine and isoproterenol, prevents the release of chemical mediators such as histamine, slow reacting substance of anaphylaxis (SRSA) and the eosinophilic chemotactic factor (ECF). A decrease in the CAMP level of the mast cell, as occurs following allergenic challenge, or increased levels of anaphylotoxin (C_3a and C_5a), as occurs in infections, will lower the level of CAMP, chemical mediators are released and asthma is produced. Epinephrine is known to increase the level of CAMP in both liver cells and mast cells. Insulin is known to lower the level of CAMP in the liver cells. Does it also do this in the mast cells? (See Figures 1 and 2) If so, it would certainly explain the aggravation of asthma in our patient by insulin. Inasmuch as the effect of insulin upon the CAMP and cyclic guanosine monophosphate (G.M.P.) level of the mast cell has never before been examined, the matter was presented to David Brennan, Ph.D., Chief of Research at Eli Lilly and Company. Various methods for assaying the cyclic noccleotides are available.[24,25] The following studies were reported by David Brennan, Ph.D. and Peter Ho, Ph.D., also of the Eli Lilly Research Department:

Histamine release from mast cells has been shown to be inhibited by elevation of endogenous CAMP. This effect was produced by the exogenous administration of epinephrine, prostaglandin E_1 and aminophylline.[26] Insulin was examined in this system for its possible opposite effect to epinephrine.

Mastocytoma cells P815 were grown in the peritoneal cavity of DBA/2 mice for eight days. The peritoneal cell suspension was washed by centrifugation and resuspension in Tyrode's medium containing 0.1% BSA (Medium A). In these experiments 39.6×10^5 mastocytoma cells in 10 ml of Medium A were added to 20 ml scintillation vials containing 0.1 ml of a solution of insulin in Medium A, or 0.1 ml of

Medium A alone as a control. Incubations were performed at 37°C with gentle shaking. At various indicated times two ml of the cell suspension were withdrawn and centrifuged at 100 x g for five minutes at 4°C. The supernatant solutions were obtained by decantation, frozen, and stored at -20°C for later histamine and CAMP determinations. Cell pellets were resuspended in 2 ml of Medium A and boiled for five minutes. The cell extract was stored at -20°C for later biochemical determination.

Histamine content was assayed by a specific enzyme-isotopic method according to the procedure of Taylor and Snyder.[27] Percentage of histamine release was calculated on the relative amount of histamine in the supernatant and in the cells. Assays of CAMP were based on the procedure of Kaliner and Austin.[26] Table IV shows that insulin at 12, 120, 1,200 and 12,000 units/ml does not increase histamine release or CAMP levels in the cells even after 60 minutes of incubation at 37°C.

According to Goodman and Gilman,[28] inhibition of insulin secretion is an alpha adrenergic response. After beta-blockade catacholamines exert only a slight stimulation of insulin secretion. They also stated that prior to insulin secretion by the pancreas increased impulse can be measured in vagal stimulation of the pancreas. This can be blocked by atropine.

If indeed insulin caused an increase in GMP or a decrease in CAMP in the mast cells, this would certainly explain the aggravation of asthma by insulin. An increase in GMP or a decrease in CAMP would result in mediator release (see Figure 2) with resultant bronchial edema and bronchospasm and a triggering or aggravation of asthma. However, Dr. Ho was unable to demonstrate such an action on rat mast cells *in vitro* by the addition of insulin. Thus, how do we explain the adverse effect which insulin produces in asthmatics?

Strom, Bear and Carpenter[29] have demonstrated that insulin increases the cyclic guanosine monophosphate (GMP) level of cells as well as enhancing the ability of cyto-toxic lymphocytes of the T system to injure target cells. Haddock et al[30] demonstrated that Propanolol, a beta-blocking agent, plus norepinephrine, as well as cholinergic stimulation with acetyl-choline increased the cellular level of cyclic GMP. Their studies reveal that cholinergic and alpha adrenergic stimulation act upon separate independent receptors. Exactly how insulin produces an increase in cyclic GMP or whether there is a separate insulin receptor is unknown. Dr. Ho was unable to find evidence of an insulin receptor on the rat mast cells *in vitro*. It would appear that the GMP increase is likely an *in vivo* reaction via vagus stimulation as well as alpha adrenergic stimulation. It may also be that rat mast cells differ from human mast cells and human basophils which future studies will clarify.

Dr. Bruce Petersen's data demonstrates some rise in anti-pork insulin IgE. Although not marked, it nonetheless is elevated. The patient was instructed to switch entirely to beef insulin to be certain we were not overlooking an atopic component in his asthma. There was no change in the clinical progress of the patient so

TABLE IV. EFFECT OF INSULIN ON HISTAMINE RELEASE AND cAMP LEVELS FROM MASTOCYTOMA CELLS.

	Histamine Release net %	cAMP pmoles
No addition	6.8	1.0
Insulin 12 unit/ml	7.6	1.1
120	5.1	1.05
1,200	7.4	0.95
12,000	6.4	1.0

that this slight IgE elevation of anti-pork insulin IgE was not felt to be a factor in his disease.

Summary

The infrequency of diabetes mellitus and asthma in the same individual is re-examined. The antagonism between epinephrine and insulin, as suggested by Konig in 1935, is indeed accurate. The assays done by the Eli Lilly Research Department revealed no *in vitro* effect of insulin on the CAMP and GMP level of mast cells as occurs in liver cells. It is felt that this effect is probably an *in vivo* effect produced via the vagus nerve and alpha-adrenergic receptor system stimulation. This would explain the mechanism of aggravation of asthma by excess insulin. Dr. Petersen's studies, the negative intradermal skin tests to insulin and the absence of change on either beef or pork insulin usage by our patient all point to a non-atopic factor in the aggravation of the asthma of this patient. In the uncommon occurrence of asthma and diabetes in the same patient, insulin dosage should be considered as a factor in all such asthmatics who do not respond well to conventional therapy. Two additional asthmatics who also have diabetes did improve with cessation of nocturnal asthma by a reduction of their evening dose of insulin.

A high fat, low carbohydrate diet, as suggested by Abrahamson to avoid dietary hyperinsulinism, is certainly worth considering in patients with nocturnal asthma. If patients cannot be made to follow a diet requiring frequent feedings high in protein and fats and low in carbohydrates, another approach suggests itself. Abrahamson was able to relieve the patients who developed nocturnal asthma with hypoglycemia by having them drink a glass of milk. Assuming other causes have been eliminated and a patient awakens each day at 3:00 a.m., an alarm clock could be set at 2:00 a.m. Milk or a milk substitute in milk sensitive patients could be taken at 2:00 a.m. to raise the blood sugar and hopefully prevent the asthma associated with hypoglycemia. Also to be noted is the ubiquitous use of tartrazine in so many drugs, including those used to relieve asthmatic symptoms.

Inasmuch as tartrazine so frequently aggravates the aspirin sensitive asthmatic, it would be wise for pharmaceutical companies to become cognizant of this fact and delete tartrazine from their antiasthmatic preparations.

Acknowledgements

The author thanks Eli Lilly and Company and especially David Brennan, Ph.D., Chief of Research, Bruce Petersen, Ph.D., and Peter Ho, Ph.D., of the Research Department of Eli Lilly and Company for their aid, without which this paper could not have been completed.

The author also thanks Eli Lilly and Company for the preparation of Figures 1 and 2 of this paper.

Edward Platcow, Ph.D., Department of Pharmacology, Eli Lilly and Company, was most helpful in researching for proper references and reviewing the text of this paper.

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"Nature gives to every time and season some beauties of its own, and from morning to night, as from the cradle to the grave, is but a succession of changes so gentle and easy that we can scarcely mark their progress."

Charles Dickens